Paper code: PHY-MS-26 Roll No. | | | | | | | | | | |

PHYSICS MID-SEMESTER EXAMINATION 2026

Mock paper for the Exam Guard demo. Not an official examination.

Time: 90 minutes Maximum marks: 30

Answer all questions. Marks for each question are shown on the right.
Use of a non-programmable calculator is allowed. Draw diagrams where they help.

1. State Newton's second law of motion and write it in terms of momentum. **[2]**

2. A car speeds up from 18 km/h to 54 km/h in 5 s. Find its acceleration in $\mathrm{m/s^2}$. **[3]**

3. A stone is dropped from a 45 m tall building. Taking $g = 9.8\ \mathrm{m/s^2}$, find the time it takes to reach the ground and its speed on impact. **[5]**

4. Define the work done by a constant force. When is the work done by a force zero? Give one example. **[4]**

5. Two resistors of 6 Ω and 3 Ω are connected in parallel across a 12 V battery. Draw the circuit and find the current drawn from the battery. **[5]**

6. Explain why a convex lens can form both real and virtual images. Use ray diagrams. **[6]**

7. A 2 kg block slides on a rough floor with an initial speed of 4 m/s and stops after 8 m. Find the coefficient of kinetic friction. **[5]**

End of paper